Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/17/2025
PI Name:	Mickens, Matthew Ph.D.		
Project Title:	Microgreens Food Safety and Antifungal Agent Analysis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Mail code: UB-A		
PI Address 2:	International Space Station Processing Facility, 3018i		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899-0001	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/01/2025	**End Date:**	03/31/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2026 per T. Sirmons/HRP. Original date was 09/30/2025 (Ed., 3/5/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Microgreens are crop seedlings that can be grown very quickly and densely to provide a flavor-packed fresh nutrient supplement and additional menu options for the crew diet. However, there are currently no means to deploy the high-quantity of seeds required to grow them safely in microgravity without flowing water, loose substrate, or excessive hardware. This investigation will analyze the flash-frozen tissue from previously harvested microgreens to measure if the Plant Preservative Mixture (PPM) antifungal agent was taken up by the crops to understand possible food safety risks.		

Rationale for HRP Directed Research:	This research is directed because NASA must define complete scientific activities in a short time and there is insufficient time to issue a solicitation.
Research Impact/Earth Benefits:	
Task Progress:	This project achieved its primary objective of determining the food safety implications of using Plant Preservative Mixture (PPM) as an antifungal additive in agar-based substrates for microgreens cultivation. Following an exhaustive search, AGQ Labs (Oxnard, CA) was selected as the analytical partner to develop and validate a method capable of detecting isothiazolinone-based active ingredient compounds for PPM in edible plant tissue. The validated method, based on QuEChERS extraction and LC-MS/MS detection, demonstrated sensitivity down to 0.05 ppm, with acceptable recovery, precision, and reproducibility parameters. Using this validated approach, twelve kale microgreen samples grown with 0.025% PPM at KSC were analyzed, and all were confirmed to be free of detectable residues. This result provides strong evidence that PPM, at the tested concentration, does not accumulate in edible microgreen tissue and can be safely used as a substrate additive to suppress and reduce fungal contamination during growth. Importantly, these findings extend beyond food safety assurance. The ability to cultivate microgreens free of PPM residues while maintaining antifungal protection strengthens the case for deploying agar-based substrates in the Microgreens Agar-based Rebar System (MARS) hardware. This hardware, developed under prior funding, provides a stable growth platform for agar and seed films, designed for use in microgravity environments. The integration of PPM into this system enables the production of single-serving, ready-to-harvest microgreens with no additional water inputs, reduced microbial risk, and minimal crew intervention. Together, the outcomes of this project demonstrate a viable pathway for producing safe, nutritious microgreens for consumption in spaceflight conditions. Ultimately, this work advances the readiness of microgreen cultivation technologies that support crew health and diet diversification on the International Space Station (ISS) and possibly future exploration missions.
Bibliography Type:	Description: (Last Updated: 02/12/2019)